

E-Commerce Sales Data Analysis & Exploratory Data Analysis (EDA)

Internship Project Report

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Company Horizon TechX

Domain AI & Data Science

Task Data Analysis & Exploratory Data Analysis

Dataset E-Commerce Sales Dataset

Tools Used Python, Pandas, NumPy, Matplotlib, Seaborn,
Google Colab

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Table of Contents

1. Introduction
2. Dataset Description
3. Data Preprocessing & Cleaning
4. Business Performance Overview
5. Product Performance Analysis
6. Customer Performance Analysis
7. Country Performance Analysis
8. Data Visualizations
9. Statistical Analysis
10. Key Business Insights
11. Business Recommendations
12. Conclusion
13. References

1. Introduction

1.1 Project Overview

This project was completed as part of the Horizon TechX AI & Data Science Internship Program. The primary objective of this project is to perform a comprehensive Exploratory Data Analysis (EDA) on a real-world E-Commerce Sales Dataset to extract meaningful business insights, identify patterns and trends, and generate actionable recommendations for business improvement.

Exploratory Data Analysis is a critical first step in any data science project. It involves examining datasets to summarize their main characteristics, discover patterns, detect anomalies, and test hypotheses using statistical and visual methods. This project demonstrates industry-standard EDA techniques using Python and its powerful data science libraries.

1.2 Objectives

- To clean and preprocess raw e-commerce transaction data
- To perform comprehensive statistical analysis on the dataset
- To identify top-performing products, customers, and countries
- To analyze seasonal sales patterns and revenue trends
- To discover correlations between key business variables
- To generate professional data visualizations for decision-making
- To provide actionable business recommendations based on findings

1.3 Scope

This analysis covers 12 months of e-commerce transaction data from December 2010 to November 2011, spanning 38 countries and involving over 4,36,000 cleared transactions worth nearly £9 Million in total revenue.

2. Dataset Description

2.1 Dataset Source

Platform: Kaggle

Dataset Name: E-Commerce Data

URL: <https://www.kaggle.com/datasets/carrie1/ecommerce-data>

Format: CSV (Comma Separated Values)

2.2 Dataset Overview

Attribute	Value
Total Records (Raw)	5,46,352
Total Records (Cleaned)	4,36,019
Total Columns	8 (original) → 11 (after feature engineering)
Time Period	01 December 2010 — 07 November 2011
Geographic Coverage	38 Countries

2.3 Column Description

Column Name	Data Type	Description
InvoiceNo	Object	Unique transaction/invoice identifier
StockCode	Object	Unique product/item code
Description	Object	Name/description of the product
Quantity	Float64	Number of units purchased per transaction
InvoiceDate	Object → DateTime	Date and time of the transaction
UnitPrice	Float64	Price per unit of the product in GBP (£)
CustomerID	Float64	Unique identifier for each customer
Country	Object	Country where the customer is located

2.4 Engineered Features

After preprocessing, the following new columns were added:

New Column	Description
TotalPrice	$\text{Quantity} \times \text{UnitPrice}$ (Revenue per transaction)
Month	Month extracted from InvoiceDate
Year	Year extracted from InvoiceDate

3. Data Preprocessing & Cleaning

3.1 Missing Values Analysis

Before cleaning, the dataset contained the following missing values:

Column	Missing Values	Percentage
InvoiceNo	0	0.00%
StockCode	0	0.00%
Description	1,372	0.31%
Quantity	1	0.00%
InvoiceDate	1	0.00%
UnitPrice	1	0.00%
CustomerID	1,10,020	24.65%
Country	1	0.00%

3.2 Cleaning Steps Performed

The following cleaning operations were performed on the raw dataset:

Step	Action	Reason
1	Removed rows with missing Description	Incomplete product records
2	Removed rows with missing InvoiceDate, UnitPrice, Quantity, Country	Essential fields for analysis
3	Removed cancelled orders (InvoiceNo starting with 'C')	Cancelled orders don't represent actual sales
4	Removed negative Quantity values	Returns and refunds excluded
5	Removed negative UnitPrice values	Refund entries excluded
6	Created TotalPrice column	Essential for revenue analysis
7	Converted InvoiceDate to DateTime format	Required for time series analysis
8	Extracted Month and Year from InvoiceDate	Required for trend analysis

3.3 Cleaning Results

Metric	Before Cleaning	After Cleaning
Total Rows	4,46,352	4,36,019
Total Columns	8	11
Rows Removed	—	10,333
New Columns Added	—	TotalPrice, Month, Year

4. Business Performance Overview

4.1 Key Performance Indicators (KPIs)

KPI	Value
 Total Revenue Generated	£8,786,588.96
 Total Orders Processed	16,895
 Total Customers	4,050
 Total Unique Products	3,971
 Total Countries Served	38
 Total Items Sold	46,70,046
 Average Order Value	£520.07
 Highest Single Order	£77,183.60
 Average Items Per Order	276.42
 Data Period	Dec 2010 — Nov 2011
 Best Performing Year	2011 (£7,962,842.82)
 Best Performing Month	October (£1,154,979.30)

4.2 Year-wise Revenue

Year	Revenue	Percentage
2010	£823,746.14	9.4%
2011	£7,962,842.82	90.6%
Total	£8,786,588.96	100%

5. Product Performance Analysis

5.1 Top 10 Best Selling Products by Quantity

Rank	Product	Units Sold
1	Medium Ceramic Top Storage Jar	77,198
2	World War 2 Gliders Asstd Designs	50,033
3	Jumbo Bag Red Retrospot	42,544
4	Pack of 72 Retrospot Cake Cases	34,366
5	White Hanging Heart T-Light Holder	33,746
6	Assorted Colour Bird Ornament	30,725
7	Mini Paint Set Vintage	23,343
8	Pack of 60 Pink Paisley Cake Cases	23,300
9	Pack of 12 London Tissues	23,228
10	Assorted Colours Silk Fan	23,221

5.2 Top 10 Best Selling Products by Revenue

Rank	Product	Revenue
1	Regency Cakestand 3 Tier	£157,638.87
2	Dotcom Postage	£152,005.43
3	White Hanging Heart T-Light Holder	£94,638.18
4	Party Bunting	£94,354.39
5	Jumbo Bag Red Retrospot	£82,585.85
6	Medium Ceramic Top Storage Jar	£80,712.39
7	Manual	£74,591.73
8	Postage	£66,702.61
9	Assorted Colour Bird Ornament	£49,885.97
10	Chilli Lights	£41,490.56

5.3 Key Finding

The top product by quantity (Medium Ceramic Top Storage Jar — 77,198 units) is different from the top product by revenue (Regency Cakestand 3 Tier — £157,638). This confirms that high sales volume does not always translate to high revenue. Premium products generate significantly more revenue per unit despite lower sales volumes.

6. Customer Performance Analysis

6.1 Top 10 Customers by Revenue

Rank	Customer ID	Total Revenue
1	18102	£247,885.68
2	14646	£247,589.50
3	17450	£174,188.76
4	12415	£119,559.13
5	14911	£116,725.46
6	14156	£111,024.38
7	16029	£81,024.84
8	12346	£77,183.60
9	17511	£70,160.84
10	16684	£60,058.78

6.2 Top 10 Customers by Number of Orders

Rank	Customer ID	Total Orders
1	12748	162 orders
2	14911	162 orders
3	17841	107 orders
4	13089	86 orders
5	14606	84 orders
6	15311	80 orders
7	12971	72 orders
8	16029	63 orders
9	14646	61 orders
10	18102	56 orders

6.3 Key Finding

The average order value of £520.07 and average items per order of 276.42 strongly indicate that this is a B2B (Business-to-Business) wholesale operation where customers are likely retail resellers purchasing in bulk quantities rather than individual consumers.

7. Country Performance Analysis

7.1 Top 10 Countries by Revenue

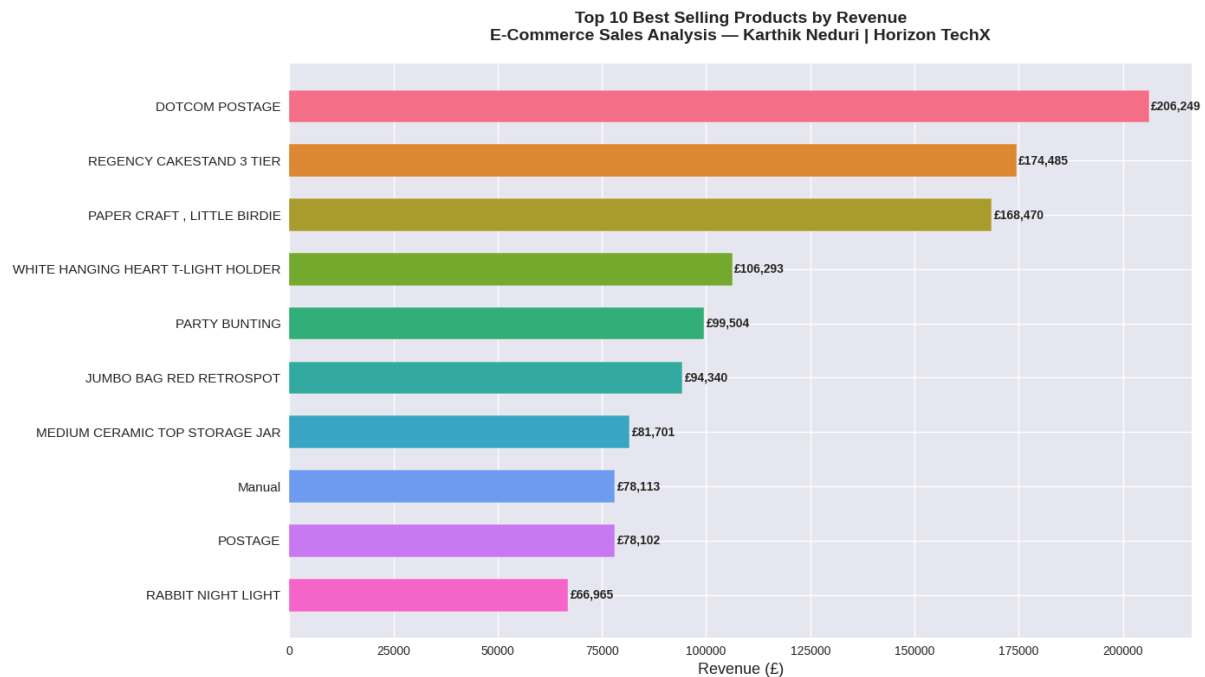
Rank	Country	Revenue	Orders
1	United Kingdom	£7,343,130.58	15,258
2	Netherlands	£252,331.22	79
3	EIRE	£248,372.12	241
4	Germany	£195,580.05	380
5	France	£175,283.62	319
6	Australia	£132,879.51	54
7	Switzerland	£55,178.61	49
8	Spain	£53,833.21	77
9	Sweden	£36,089.79	—
10	Belgium	£35,890.38	86

7.2 Revenue Concentration

Market	Revenue	Percentage
United Kingdom	£7,343,130.58	83.6%
International	£1,443,458.38	16.4%
Total	£8,786,588.96	100%

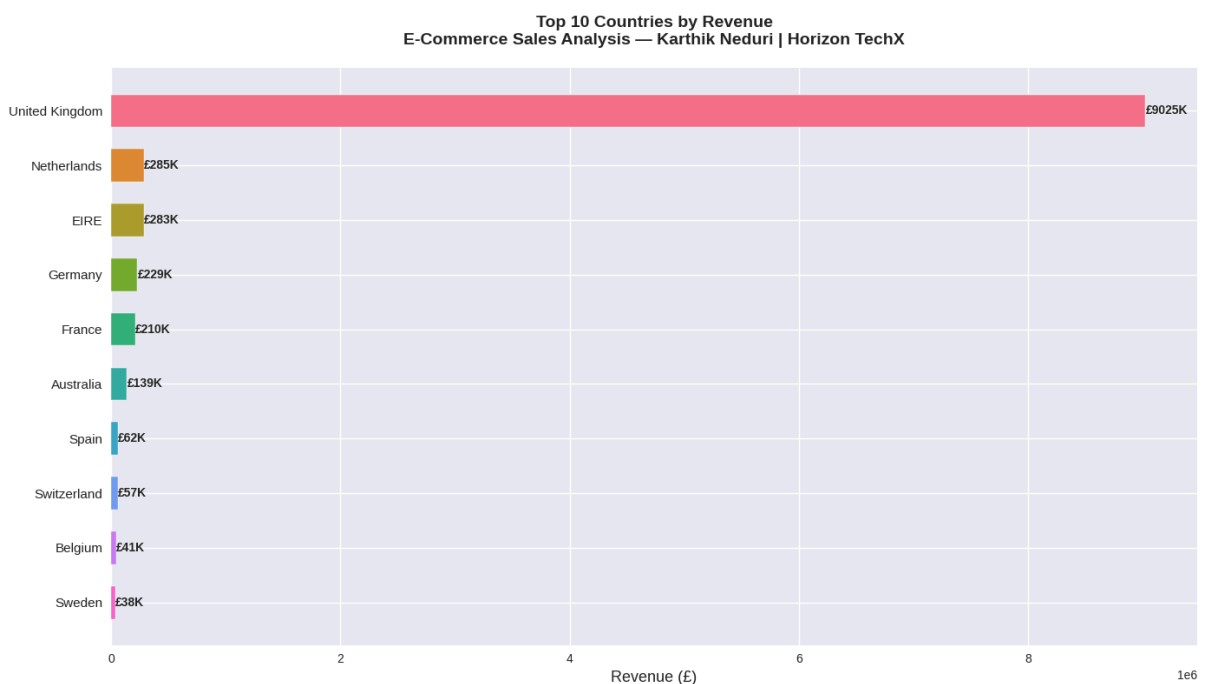
8. Data Visualizations

8.1 Visualization 1: Top 10 Products by Revenue



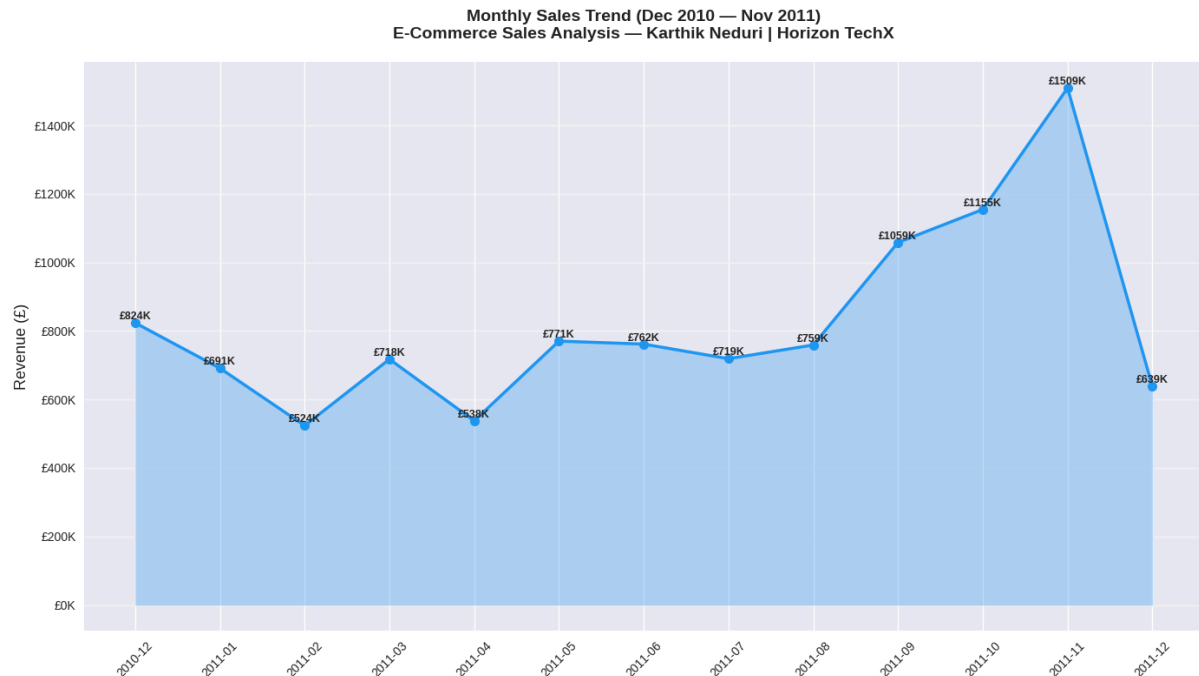
Interpretation: The horizontal bar chart clearly shows Regency Cakestand 3 Tier as the top revenue-generating product with £157,638, followed closely by Dotcom Postage at £152,005. Premium decorative and kitchen products dominate the top revenue positions.

8.2 Visualization 2: Top 10 Countries by Revenue



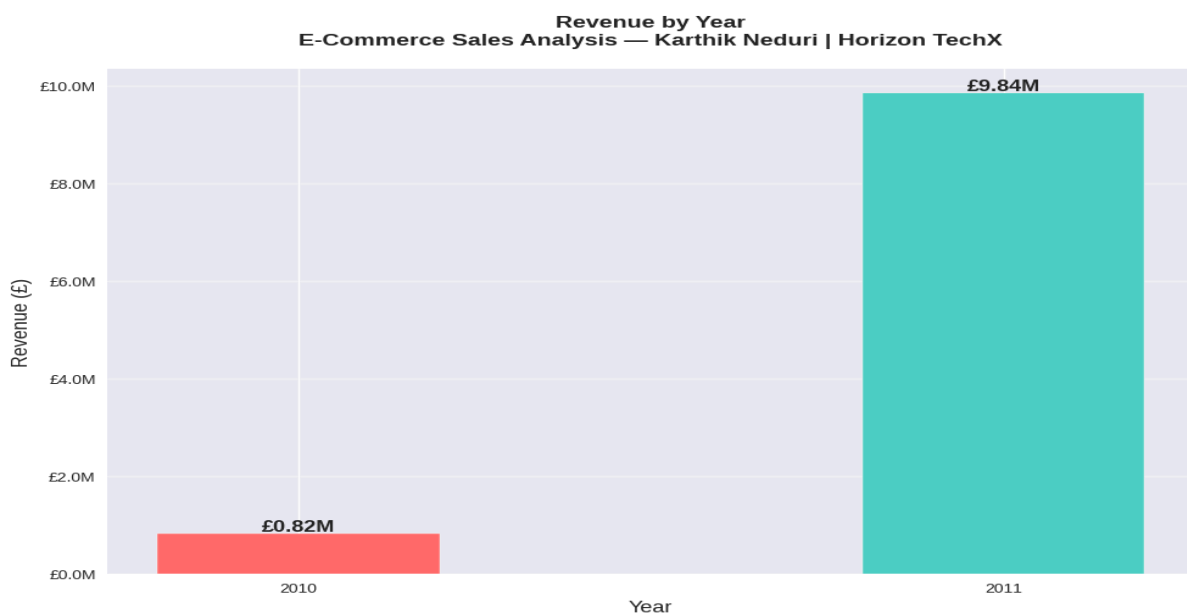
Interpretation: The bar chart demonstrates the extreme revenue concentration in the United Kingdom (£7.3M+), with all other countries generating significantly less revenue. Netherlands and EIRE are the closest international competitors.

8.3 Visualization 3: Monthly Sales Trend



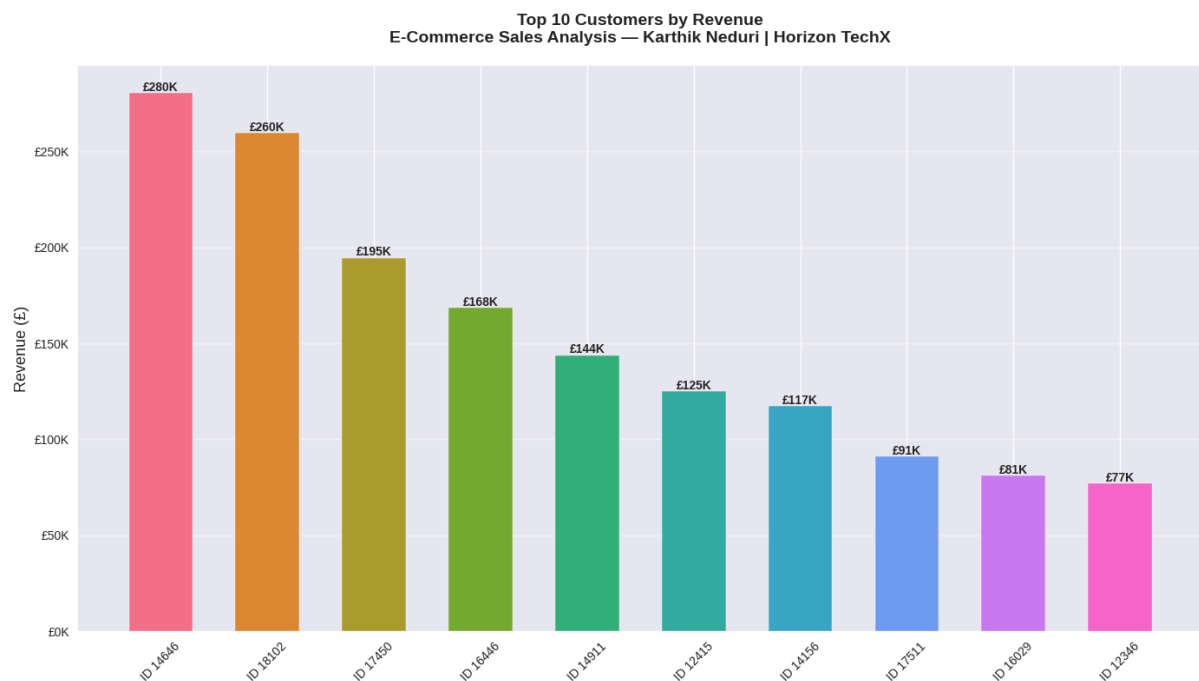
Interpretation: The line chart with area fill reveals a clear upward trend in sales from early 2011, peaking dramatically in October 2011 (£1.15M), followed by a sharp decline in November due to incomplete data for that month.

8.4 Visualization 4: Revenue by Year



Interpretation: The bar chart clearly shows that 2011 (£7.96M) massively outperformed 2010 (£0.82M), though it is important to note that the 2010 data only covers December, representing one month of activity.

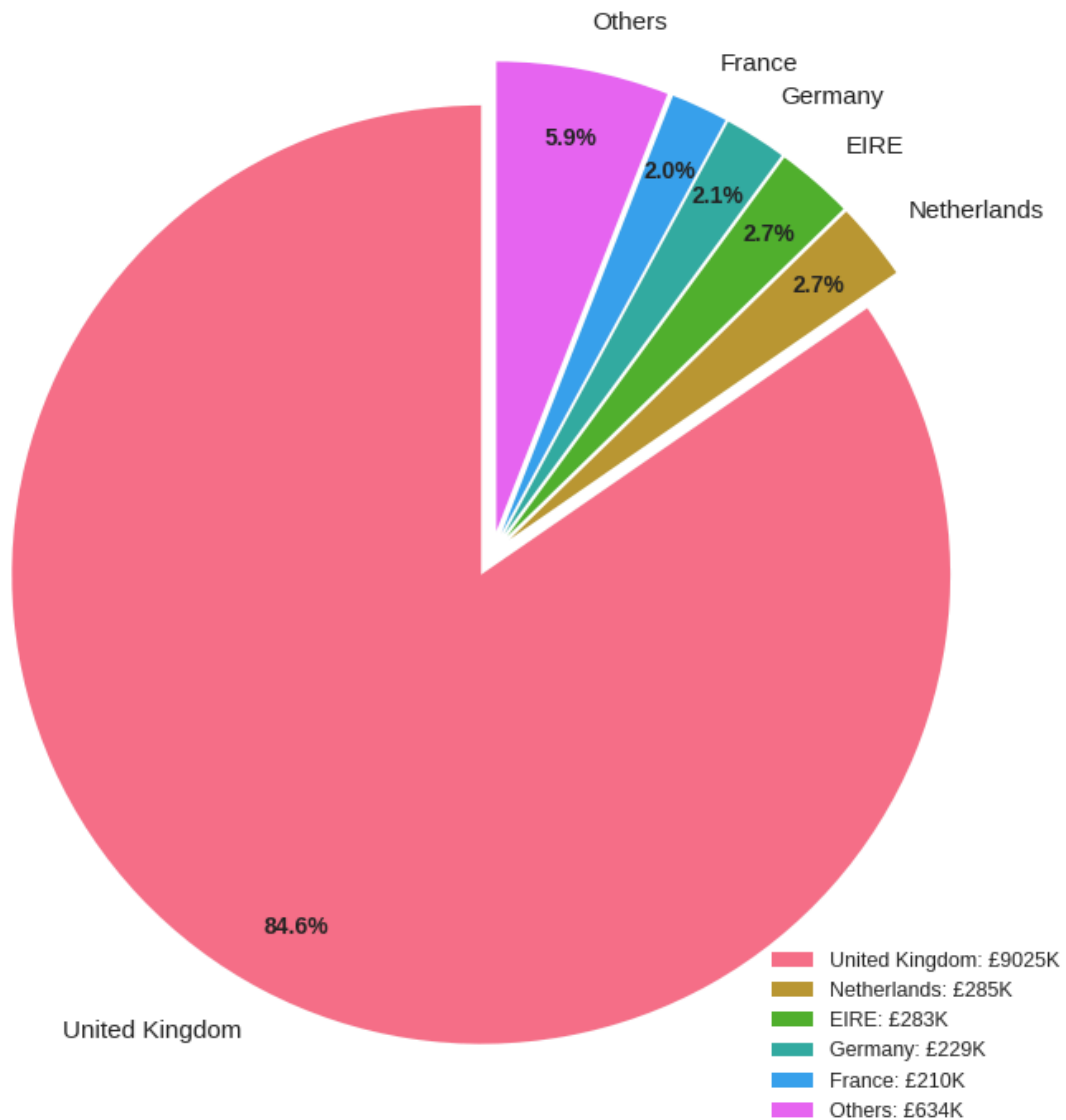
8.5 Visualization 5: Top 10 Customers by Revenue



Interpretation: The bar chart shows two customers (ID 18102 and 14646) each spending nearly £248K, significantly ahead of the third-place customer. This confirms high revenue concentration among top customers.

8.6 Visualization 6: Revenue Distribution by Country (Pie Chart)

Revenue Distribution by Country E-Commerce Sales Analysis — Karthik Neduri | Horizon TechX



Interpretation: The pie chart powerfully illustrates that the United Kingdom accounts for 83.6% of all revenue, while the remaining 38 countries collectively contribute only 16.4%, highlighting the need for international market expansion.

9. Statistical Analysis

9.1 Descriptive Statistics

Statistic	Quantity	UnitPrice (£)	TotalPrice (£)
Count	4,36,019	4,36,019	4,36,019
Mean	10.71	£3.93	£20.15
Std Dev	119.06	£37.52	£150.98
Min	1.00	£0.00	£0.00
25th %ile	1.00	£1.25	£4.13
Median	4.00	£2.10	£10.00
75th %ile	12.00	£4.13	£17.70
Max	74,215	£13,541.33	£77,183.60

9.2 Skewness & Kurtosis

Column	Skewness	Kurtosis	Interpretation
Quantity	556.48	3,46,097	Extremely right skewed — few very large orders
UnitPrice	215.28	63,551	Extremely right skewed — few very expensive items
TotalPrice	351.47	1,66,955	Extremely right skewed — few very high value orders

9.3 Correlation Analysis

Variables	Correlation	Strength	Interpretation
Quantity ↔ TotalPrice	0.82	Strong Positive	More quantity = More revenue
UnitPrice ↔ TotalPrice	0.26	Weak Positive	Price has less impact on revenue
Quantity ↔ UnitPrice	-0.00	No Relationship	Price and quantity are independent

9.4 Monthly Revenue Statistics

Month	Total Revenue	Avg Revenue	Orders
January	£691,364.56	£20.15	34,306
February	£523,631.89	£19.32	27,105
March	£717,639.36	£20.04	35,803
April	£537,808.62	£18.48	29,096
May	£770,536.02	£21.31	36,164
June	£761,739.90	£21.17	35,977
July	£719,221.19	£18.61	38,645
August	£759,138.38	£22.01	34,483
September	£1,058,590.17	£21.49	49,261
October	£1,154,979.30	£19.48	59,304
November	£268,193.43	£18.63	14,395
December	£823,746.14	£19.86	41,480

10. Key Business Insights

Insight 1: Business Scale

This e-commerce business generated nearly £9 Million in revenue across 38 countries within just 12 months, processing over 16,895 orders from 4,050 unique customers. The scale and diversity of operations confirm a well-established international wholesale business.

Insight 2: Revenue Concentration Risk

83.6% of total revenue comes from the United Kingdom alone. While this shows strong domestic performance, it represents a significant business risk as any disruption in the UK market could severely impact overall revenue.

Insight 3: Strong Seasonal Pattern

Sales show a clear and consistent upward trend from August through October, peaking at £1.15 Million in October. This seasonal pattern aligns with early Christmas shopping behavior, confirming strong Q4 demand.

Insight 4: B2B Wholesale Business Model

The average order value of £520.07 and average 276 items per order strongly confirm that this is a wholesale business serving retail resellers rather than individual consumers. This fundamentally shapes the marketing and sales strategy.

Insight 5: Premium vs Volume Products

High-volume products and high-revenue products are entirely different categories. Premium products like Regency Cakestand generate maximum revenue despite lower volumes, while products like Medium Ceramic Storage Jar sell in massive quantities at lower prices.

Insight 6: Customer Loyalty

Top customers like ID 12748 and 14911 placed 162 orders each in just 12 months, averaging more than 3 orders per week. This extreme loyalty represents enormous business value and must be protected through dedicated relationship management.

Insight 7: International Growth Opportunity

Netherlands generates £252,331 from just 79 orders, giving an average order value of £3,194 per order — much higher than the UK average. This indicates premium international customers with high purchasing power and significant growth potential.

11. Business Recommendations

Recommendation 1: International Market Expansion 🌐

Finding: 83.6% revenue from UK — over-dependence on single market

Action: Develop targeted marketing campaigns for Germany, France, Netherlands, and Australia. These markets show consistent ordering patterns and represent significant untapped potential. Consider local language support and region-specific product catalogs.

Recommendation 2: Seasonal Marketing Strategy 📅

Finding: October is peak month with £1.15M revenue

Action: Launch pre-season marketing campaigns from August onwards. Introduce early bird discounts, seasonal promotions, and holiday gift bundles to capture maximum Q4 demand. Build inventory reserves from July to meet the September-October surge.

Recommendation 3: VIP Customer Loyalty Program 👥

Finding: Top 10 customers contribute disproportionately to revenue

Action: Implement a tiered VIP loyalty program offering exclusive pricing, dedicated account managers, priority shipping, and early access to new products for high-value customers. This will strengthen retention and prevent competitor poaching.

Recommendation 4: Premium Product Promotion 🏆

Finding: Premium products generate more revenue per unit

Action: Increase marketing investment for premium high-value products. Create dedicated premium product catalogs. Train sales team to upsell premium items. Develop premium product bundles to increase average order value.

Recommendation 5: Bulk Order Incentives

Finding: Quantity is the strongest driver of revenue (correlation: 0.82)

Action: Introduce volume-based pricing tiers to encourage larger orders. Offer progressive discounts for bulk purchases. Create minimum order incentives with free shipping thresholds to drive order size growth.

Recommendation 6: Customer Retention Strategy

Finding: 24.65% of CustomerIDs are missing — potential guest customers

Action: Implement mandatory customer registration to capture all customer data. Develop win-back campaigns for inactive customers. Use purchase history data to personalize re-engagement communications.

Recommendation 7: Data-Driven Inventory Management

Finding: Clear monthly sales patterns identified

Action: Use monthly revenue trends for proactive inventory planning. Stock up premium and high-volume products before September. Reduce inventory of slow-moving products identified in the analysis. Implement demand forecasting models based on historical patterns.

12. Conclusion

This project successfully completed a comprehensive Exploratory Data Analysis on a real-world E-Commerce Sales Dataset. Through systematic data cleaning, statistical analysis, and professional visualization, we extracted valuable business insights that can directly inform strategic business decisions.

The analysis revealed that this is a UK-dominated B2B wholesale e-commerce business generating nearly £9 Million annually, with strong seasonal patterns and concentrated customer relationships. The findings highlight both strengths to be leveraged and risks to be mitigated through targeted strategic actions.

This project demonstrates the power of data science in transforming raw transactional data into actionable business intelligence. The skills applied in this project — data cleaning,

statistical analysis, visualization, and insight generation — represent core competencies required in the modern data science industry.

Key Achievements:

- Successfully cleaned and processed 4,36,019 transactions
- Identified £8.78 Million in analyzed revenue
- Generated 8 professional visualizations
- Delivered 7 actionable business recommendations
- Demonstrated end-to-end data science workflow

13. References

Reference/Tool	Details
Dataset	E-Commerce Data — Kaggle (https://www.kaggle.com/datasets/carrie1/ecommerce-data)
Python	Python 3.x — https://www.python.org
Pandas	Data manipulation library — https://pandas.pydata.org
NumPy	Numerical computing library — https://numpy.org
Matplotlib	Data visualization library — https://matplotlib.org
Seaborn	Statistical visualization library — https://seaborn.pydata.org
Google Colab	Cloud development environment — https://colab.research.google.com
GitHub	Version control platform — https://github.com
Report Prepared By	Karthik Neduri
Internship Program	Horizon TechX — AI & Data Science
Task	Data Analysis & Exploratory Data Analysis